

Experiment No-3

Student Name: Harsh

Branch: MCA

Semester: 3rd

Subject Name: Business Analytics

UID: 24MCA20045

Section/Group: 24MCA-1(B)

Date of Performance: 21/09/2025

Subject Code: 24CAH-703

Aim: This experiment aims to help students install Oracle Database and SQL Developer on their systems, configure the environment, and verify the installation by running basic SQL commands. By the end of this exercise, students will have a working Oracle database environment ready for learning SQL.

Objective of the practical:

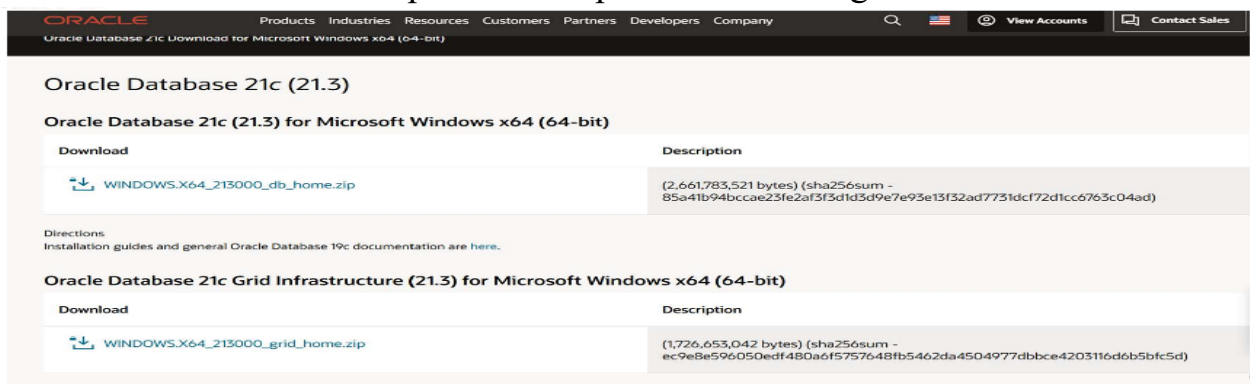
The objective of this experiment is to enable students to successfully install and configure Oracle Database and SQL Developer, establish a working database environment, and validate the setup by executing basic SQL commands. This will prepare students with the foundational environment required for practicing and learning SQL.

Steps:

Step1: Download Oracle Database

1. Open your browser and go to: <https://www.oracle.com/database>
2. Scroll down to Oracle Database 21c Express Edition (XE).
3. Select the appropriate version for your operating system (Windows 10/11 (64-bit) or Linux 64-bit).
4. Accept the Oracle License Agreement.

Click Download and wait for the setup file to complete downloading.

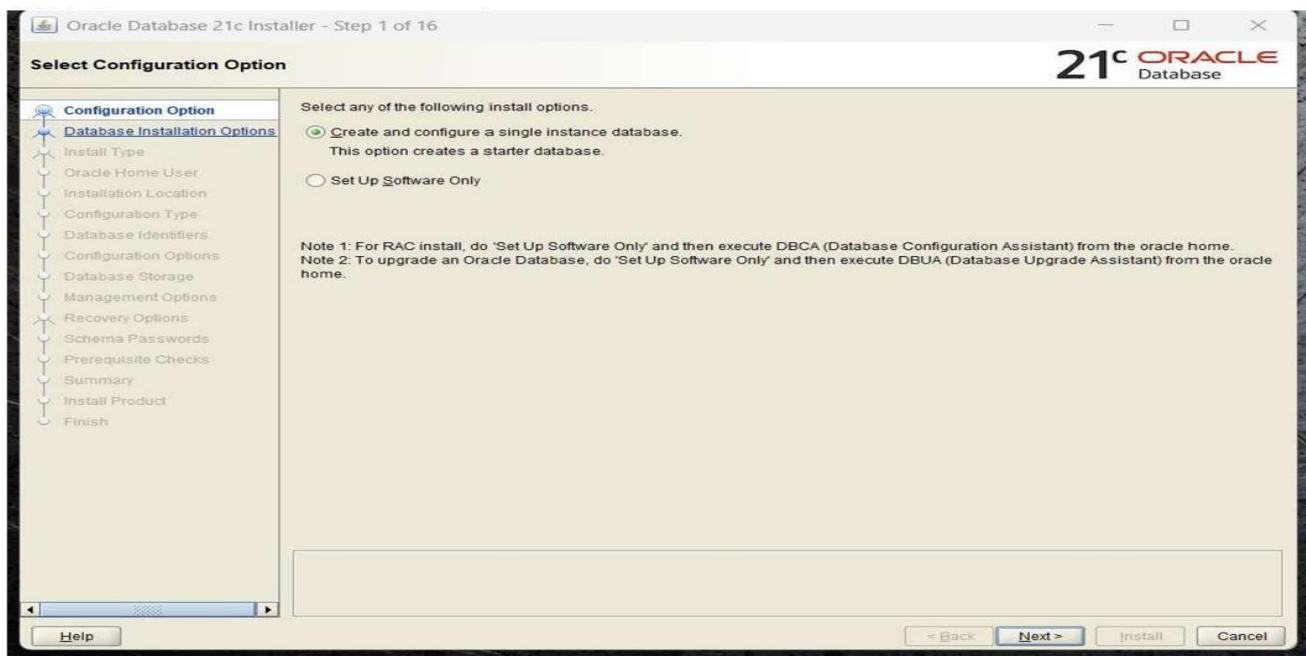


The screenshot shows the Oracle Database 21c download page. The header includes the Oracle logo and navigation links: Products, Industries, Resources, Customers, Partners, Developers, and Company. There are also links for View Accounts and Contact Sales. The main content area is titled "Oracle Database 21c (21.3)" and "Oracle Database 21c (21.3) for Microsoft Windows x64 (64-bit)". It features a table with download links and descriptions. The first row is for "WINDOWS.X64_213000_db_home.zip" with a description of (2,661,783,521 bytes) and a SHA256sum. The second row is for "WINDOWS.X64_213000_grid_home.zip" with a description of (1,726,653,042 bytes) and a SHA256sum. Below the table, there are links for "Directions" and "Installation guides and general Oracle Database 19c documentation are here."

Download	Description
WINDOWS.X64_213000_db_home.zip	(2,661,783,521 bytes) (sha256sum - 85a41b94bccae23fe2af3f3d1d3d9e7e93e13f32ad7731dcf72d1cc6763c04ad)
WINDOWS.X64_213000_grid_home.zip	(1,726,653,042 bytes) (sha256sum - ec9e8e596050edf480a6f5757648fb5462da4504977dbbce4203116d6b5bfc5d)

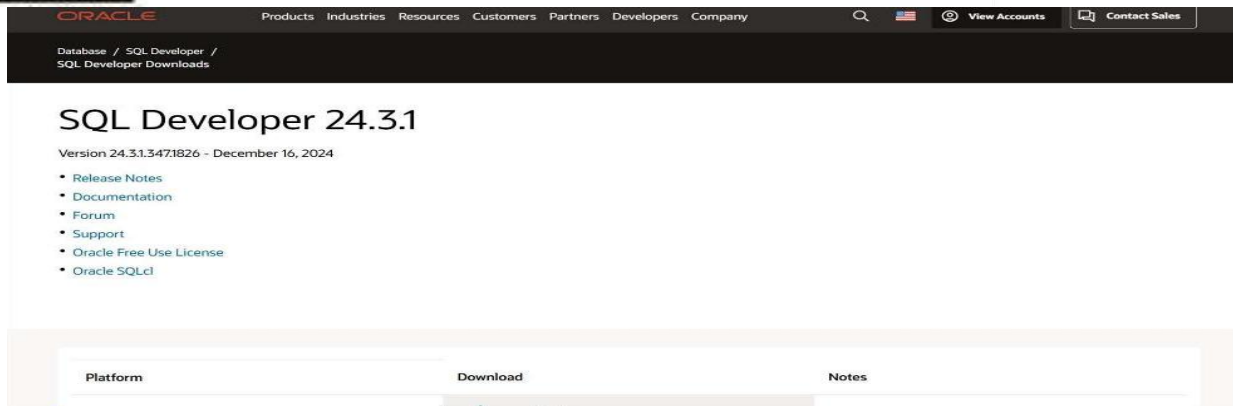
Step2: Install Oracle Database

1. Locate the downloaded installer file.
2. Right-click → Run as Administrator.
3. The installation wizard will appear → Click Next.
4. Choose the default installation settings (recommended for learning).
5. When prompted, set a strong password for SYS and SYSTEM accounts (e.g., Admin123).
6. Complete the installation (may take 15–30 minutes).
7. Once installed, Oracle services will start automatically.



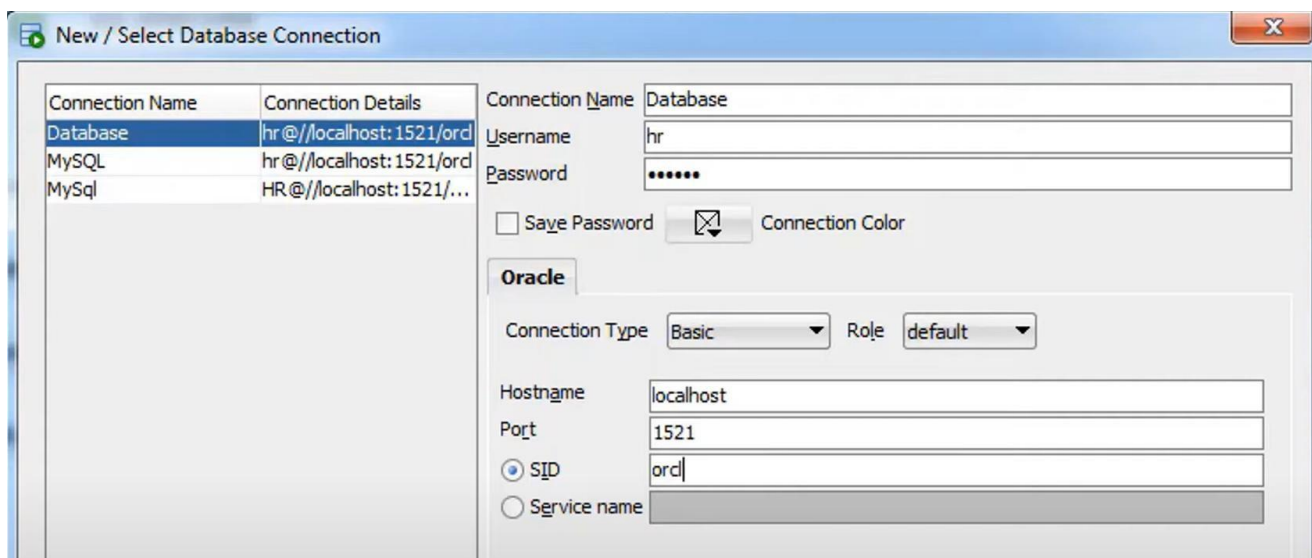
Step3: Install SQL Developer

1. Go to: <https://www.oracle.com/tools/downloads/sqldev-downloads.html>.
2. Download the latest version of Oracle SQL Developer.
3. Extract (Unzip) the downloaded file into your preferred folder.
4. Open the folder and double-click sqldeveloper.exe to launch.
 - No installation is required.



Step4: Configure Database Connection in SQL Developer

1. Open SQL Developer.
2. Click the green “+” (New Connection) icon.
3. Enter the following details:
 - Connection Name: LocalOracle
 - Username: system
 - Password: (the password you set during installation)
 - Hostname: localhost
 - Port: 1521
 - SID: XE
4. Click Test → If “Success” appears, click Save.



Step5: Verify Installation

1. In SQL Developer, connect to the database using the saved connection.
2. Open a new SQL Worksheet (File > New > SQL File).
3. Run the following SQL commands one by one:

```
SELECT * FROM v$version;  
CREATE TABLE test (id NUMBER, name VARCHAR2(50));  
INSERT INTO test VALUES (1, 'Oracle Test'); SELECT * FROM test;
```

4. If these queries run successfully and return results, your installation is verified.

```
Copyright (c) 1982, 2021, Oracle. All rights reserved.  
Enter user-name: system  
Enter password:  
Last Successful login time: Wed Aug 13 2025 14:54:40 +05:30  
  
Connected to:  
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production  
Version 21.3.0.0.0  
  
SQL> SELECT * FROM v$version;  
  
BANNER  
-----  
BANNER_FULL  
-----  
BANNER_LEGACY  
-----  
CON_ID  
-----  
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production  
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production  
Version 21.3.0.0.0  
Oracle Database 21c Enterprise Edition Release 21.0.0.0.0 - Production  
0  
  
BANNER  
-----  
BANNER_FULL  
-----  
BANNER_LEGACY  
-----  
CON_ID  
-----  
  
SQL>  
SQL> CREATE TABLE test (id NUMBER, name VARCHAR2(50));  
  
Table created.  
  
SQL>  
SQL> INSERT INTO test VALUES (1, 'Oracle Test');  
  
1 row created.  
  
SQL>  
SQL> SELECT * FROM test;  
  
ID NAME  
-----  
1 Oracle Test  
  
SQL> |
```

Example of create Table:

```
SQL> create table laptops(Name varchar(25), Model_Name varchar(25));
```

```
Table created.
```

```
SQL> select * from laptops;
```

NAME	MODEL_NAME
Acer	Aspire Lite
Dell	Inspiron 3000
HP	Intel Core i5

Learning Outcomes:

1. Gained knowledge of the core components of Oracle Database and their purpose in database management.
2. Successfully learned how to obtain the correct version of Oracle Database from the official website and complete installation.
3. Understood how to set up a local Oracle Database instance with proper connection parameters.
4. Learned to install SQL Developer and set it up for managing and querying databases.